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Accelerating Molecular Generation through Representation Alignment

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Abstract

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Chapter 1

Introduction

The evolution of generative machine learning in scientific domains is defined by the tension between data scale, data quality, and structural priors. We examine a recent addition to this story, *representation alignment*, in two regimes for de novo molecular generation — small molecules and protein backbones — and characterise its behaviour. In this introduction, we trace the narrative arc of research trends in molecular generation, and highlight the motivation for our work: the emerging need for data-efficient modelling in a world of prior-light models.

1.1 A history of trade-offs

The dominant ideas in generative modelling are *denoising diffusion* and its offshoot *flow matching* [1, 2], which found breakthrough success in the realm of image generation. This paradigm has since been adapted for de novo design challenges across scientific fields, such as generating protein backbones [17, 18, 13], inorganic materials [20], and drug-like small molecules [14]. Training these models is typically expensive in both data and compute, with state-of-the-art image generators routinely training on billions of examples [27]. However, the corpus of available high-quality molecular data sits orders of magnitude below this. Experimental determination is slow and expensive, which means datasets are correspondingly small. For example, protein structure generators have largely been trained on at most half a million structures [13], which may explain why they have not yet matched the maturity of their vision-domain counterparts.

Historically, researchers compensated for the small scale of high-quality molecular data by encoding structural priors directly into model architectures [33]. SE(3)-equivariant architectures, for example, enforce the rotational and translational symmetries expected of molecules by construction [35]. But hard structural priors come with a computational cost, and the operations required to encode geometry are a bottleneck to scaling models.

A recent wave of research has moved entirely the other way: drop hard priors, use bare-

bones non-equivariant transformer stacks, and rely on data scale and augmentation to teach the model what equivariance buys. This shift fits a broader trend in machine learning on leveraging representation instead of model architecture to inject soft inductive biases for structured data. Vision Transformers [36] showed that the spatial-locality and translation-equivariance priors of convolutional neural nets (CNNs) can be learned rather than imposed, given sufficient scale. Graph Transformers like TokenGT [37] have made the same case for graphs. AlphaFold 3 [39] stripped many of the SE(3)-equivariant biases of AlphaFold 2 [38], replacing the structure module with a diffusion model operating on raw coordinates. Brehmer et al. [40] make the empirical scaling case directly: more data through a simpler model beats less data through a richer one. In our work, we examine Tabasco [14] and Proteina [13], which are flow-based generators built on largely vanilla transformers for small molecules and protein backbones respectively.

Given the difficulty of procuring data in scientific domains, data scale must necessarily come from non-experimental sources, which may be of lower quality. For example, recent efforts in protein generation have leaned on synthetic structures from folding-model predictions [13, 29] to expand training datasets into the tens of millions. However, even these enhanced corpora are arguably insufficient. Metagenomic databases catalogue over two billion natural protein sequences [31], and the theoretical sequence space at a median protein length [32] is hundreds of orders of magnitude greater still. Even at the frontier of the field’s data scale, we sample but a sliver of the distribution we wish to model.

In summary, the field has traded hard structural priors for architectural simplicity, and curated data for synthetic scale. However, this trade-off dilutes the structural signal available to a model. Furthermore, even our best attempts at creating large datasets may not be large enough. These twin challenges amplify the need for techniques that make training still more efficient without sacrificing quality.

1.2 Representation alignment

But why is training diffusion-transformer architectures expensive in the first place? One answer comes from Yu et al. [5], who argue that these models suffer from a *representation bottleneck*. During training, a model is implicitly asked to learn two distinct tasks simultaneously: (1) representation: extracting semantically meaningful encodings from complex data; and (2) generation: constructing data from these encodings. The standard denoising objective alone may not provide sufficient signal for representation learning, leading to slow convergence and the large training budgets these models require in practice. In the context of molecular generation, this representation burden is compounded by the twin challenges outlined above. Models are now forced to encode complex structural geometry from datasets that only sparsely sample the true underlying distribution, all while operating without the aid of architectural priors.

Yu et al. propose Representation Alignment (REPA) as a remedy. The technique adds a regularisation term that aligns a generator’s intermediate hidden states with the embeddings of a frozen pretrained encoder. On image diffusion, they report roughly a 17.5× training speed-up against a strong baseline. REPA has since been extended beyond images to video [9], audio [10], and recently to molecular generation [7, 8], where it has been demonstrated but not systematically characterised.

1.3 Contributions

In this light, the core question we investigate in this report is: *can prior-light molecular generation models benefit from representation alignment with pre-trained models, in training efficiency and generation quality?* We aim to characterise when and why this works, including failure modes and limitations.

We document two lines of enquiry.

- (i) *Does REPA work in the molecular domain?*: We conduct empirical studies on training flow-based generators with REPA in two regimes: Tabasco [14] for small molecules, and Proteina [13] for protein backbones. We find that REPA does not measurably improve Tabasco, but it improves training efficiency and advances the generation quality envelope for Proteina.
- (ii) *When does REPA work?*: Following Singh et al. [6] in the realm of image generation, we develop a profiling framework to characterise the suitability of an encoder as a representation target. We measure three properties of an encoder: (1) an upper bound on the structural information it contains (linear-probe recoverability against domain structural labels); (2) a lower bound on what a projector can transfer from that information (projector-based recoverability); (3) the geometric conditioning of its embeddings for the alignment objective (effective rank).

We do not propose a new architecture or alignment loss, and our two studies are not a controlled experiment over the conditions that govern REPA’s effectiveness. Instead, they are best read as paired case studies in regimes where those conditions differ.

The remainder of this report is organised as follows. Chapter 2 sets out the background and context needed to read the rest: flow matching, REPA, molecular generation. Chapter 3 discusses what it means for a generative model to be “good” in our domain, and the tensions that question hides. Chapter 4 develops the encoder-profiling framework and uses it to predict where REPA may and may not help. Chapters 5 and 6 present the two paired studies, small molecules first and protein backbones second. Finally, we conclude in Chapter 7 with a discussion of whether what we learn generalises beyond molecular generation.

Chapter 2

Background and Related Work

In this chapter, we introduce the machinery the rest of this thesis depends on. We outline flow matching (§2.1), the generative paradigm underlying the models we study, and representation alignment (§2.2), the regulariser that is our object of study. We then turn to the molecular setting: we situate generative models within the broader de novo molecular design pipeline (§2.3), before presenting the two molecular domains we work in, small molecules and protein backbones, and the flow-matching models we extend, Tabasco and Proteina (§2.4). The treatment is cursory by design. We intend only to guide the reader through this thesis, and we provide citations for omitted details.

2.1 Flow matching

At first glance, the term *generative modelling* implies the problem of *creating* new artefacts from scratch. However, it is perhaps more accurately described as the problem of *sampling* new artefacts from a complex data distribution that can only be accessed through a finite training set. A common strategy here is to fix a simple distribution we can sample from cheaply — typically standard Gaussian noise $p_0 = \mathcal{N}(0, I)$ — and learn a mapping from it to the desired data distribution p_1 .

Flow matching (FM) realises this mapping as a continuous-time transport. The model designer defines a *probability path* between a noise sample and a data sample — a continuous interpolation between them — and the model learns to carry the noise sample along this path. Concretely, it parameterises a time-dependent *velocity field* $v_\theta(x, t)$ that gives the instantaneous direction of motion at each location x and time $t \in [0, 1]$. At inference, integrating the learned ordinary differential equation (ODE) $\dot{x}_t = v_\theta(x_t, t)$ from $t = 0$ to $t = 1$ pushes a fresh noise sample forward into a fresh data sample.

A common choice of probability path is the linear interpolant, $x_t = (1 - t)x_0 + tx_1$ for $x_0 \sim p_0$ and $x_1 \sim p_1$. The model’s target is the *marginal* velocity field at (x, t) : the average direction of motion over all noise-data pairs whose paths pass through that point.

However, supervising this marginal directly is intractable, since the contributing pairs are not known in advance. Flow matching sidesteps this difficulty with a key observation [2]: although the marginal field may be intractable, the *conditional* velocity along the path of any specific noise-data pair is relatively trivial. Indeed, for the linear interpolant, it is simply $\dot{x}_t = x_1 - x_0$. Regressing the model on these conditional velocities, averaged over many sampled pairs, recovers the marginal field in expectation, and gives the training loss as a mean-squared regression:

$$\mathcal{L}_{\text{FM}} = \mathbb{E}_{t, x_0, x_1} \|v_\theta(x_t, t) - (x_1 - x_0)\|^2.$$

In practice, every training step samples a time $t \in [0, 1]$ and a noise-data pair (x_0, x_1) , computes the interpolant x_t , and takes a gradient step on this squared error.

Flow matching with this linear interpolant generalises score-based diffusion: the variance-preserving Gaussian-path objective of denoising diffusion models [1] is a special case of the FM family [2, 4].

Generating samples from a trained flow-matching model amounts to integrating the learned velocity field. This is often performed with a standard ODE solver, with Euler steps usually sufficing for stable trajectories. The field also admits stochastic sampling variants — such as those drawing on the connection to score-based stochastic differential equations (SDEs) — which inject noise during the integration steps to explore the probability space and improve sample diversity. The models we work with in this report, Tabasco and Proteina, support both deterministic and stochastic samplers.

Conditional generation augments v_θ with context tokens — such as a class label, a sequence length, or a structural motif — with conditioning supplied at both training and sampling time. In this report, we focus on unconditional generation only.

For a more complete treatment of flow matching, we refer the reader to Lipman et al. [2] for the original presentation, and to the recent Lipman et al. guide [4].

2.2 Representation alignment

Chapter 1 introduced the argument made by Yu et al. [5] about the *representation bottleneck* in diffusion-transformer models. Their analysis applies to flow-matching models as well, given the equivalence between the two training objectives discussed above (§2.1). The flow-matching objective also implicitly asks a model to learn two things at once: a useful representation of the data, and a noise-to-data velocity field that interprets this representation. This twin burden, they argue, contributes to slow convergence and large training budgets.

Their proposed solution, Representation Alignment (REPA), addresses the first half of this burden by adding a regularisation term to the training loss. Figure 2.1 shows the

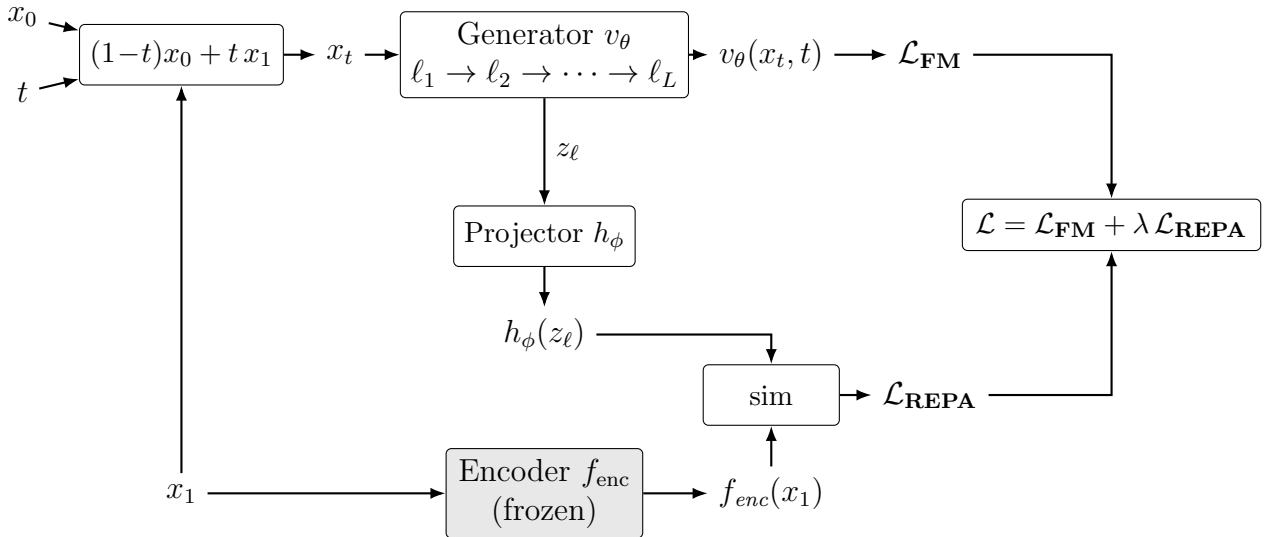


Figure 2.1: The REPA construction. A noised sample $x_t = (1-t)x_0 + tx_1$ is constructed at each training step from a noise sample x_0 , a clean data sample x_1 , and a sampled timestep $t \in [0, 1]$. The generator v_θ processes x_t through its sequence of layers $\ell_1, \dots, \ell_L$ to produce the velocity field that feeds the flow-matching loss $\mathcal{L}_{\text{FM}}$. Separately, the clean sample x_1 is also passed through a frozen pretrained encoder f_{enc} (shaded). The generator’s hidden state z_ℓ at the chosen alignment layer is mapped into the encoder’s feature space by a trainable projector h_ϕ ; the REPA loss $\mathcal{L}_{\text{REPA}}$ is the negated similarity between $h_\phi(z_\ell)$ and $f_{\text{enc}}(x_1)$, averaged across tokens. The total training loss $\mathcal{L}$ combines them with weight λ .

construction. The regularisation pulls z_ℓ , the generator’s intermediate hidden state at a chosen *alignment layer* ℓ , towards the embeddings produced by a frozen pretrained encoder f_{enc} run on the clean data sample x_1 . The hidden state z_ℓ is a sequence of N token vectors. In the context of images, each token represents an image patch; in the molecular setting, each token represents an atom or residue that is part of the larger molecule (§2.3). A small trainable *projector head* h_ϕ — a three-layer multi-layer perceptron (MLP) in the original paper — maps z_ℓ into the encoder’s feature space, so that $h_\phi(z_\ell)$ and $f_{\text{enc}}(x_1)$ can be compared. This alignment is asymmetric: the encoder provides a fixed target throughout training, while only the generator parameters θ and the projector ϕ are updated.

The alignment loss is the negated similarity, averaged token-wise between projected and target embeddings:

$$\mathcal{L}_{\text{REPA}} = -\mathbb{E}_{t, x_0, x_1} \left[\frac{1}{N} \sum_{n=1}^N \text{sim}(h_\phi(z_\ell)_n, f_{\text{enc}}(x_1)_n) \right],$$

where n indexes the N tokens within a sample. In the image setting, N is constant across the dataset, so this token-wise average is equivalent to averaging similarities once per sample and then across the batch — the original paper draws no distinction between the two. For molecular data, where N varies per sample, the two diverge. We return to this distinction in the design knobs below.

The final training objective is the flow-matching loss from §2.1 plus this term, scaled by a weight λ :

$$\mathcal{L} = \mathcal{L}_{\text{FM}} + \lambda \mathcal{L}_{\text{REPA}}.$$

This construction exposes several design choices, some of whose effects we revisit in the ablations of Chapter 6:

- The choice of encoder f_{enc} . Different encoders may capture distinct aspects of the data distribution.
- The alignment layer ℓ at which we attach the projector. Yu et al. experimentally determined that mid-network layers work best for image diffusion, but the optimal choice is not obvious a priori.
- The alignment-loss weight λ , which balances the alignment signal against the generative objective. Once again, Yu et al. experimentally determined an optimal value of 0.5 for image diffusion, but this may not hold universally.
- The choice of similarity metric sim . Yu et al. compared cosine similarity and mean-squared error for image diffusion, and found the former more effective. We adopt cosine similarity in this report as well.
- The averaging mode. Averaging tokens within each sample first weights every molecule equally, while pooling tokens across the batch gives larger molecules proportionally more REPA signal.

We adopt the core REPA construction as in Yu et al. for this report. However, a small family of methodological variants on REPA has since emerged. Dispersive regularisers [12] replace the external encoder with an internal contrastive term, while flexible-target formulations [7] broaden which encoder features the loss attaches to. Separately, REPA-style alignment has been applied beyond the image setting, to video [9], audio [10], materials [11], and molecules [8]. We defer exploration of these variants to future work.

2.2.1 What makes for a good REPA encoder?

There are many possible choices for the encoder f_{enc} in any REPA construction, and this selection is a key design decision. But what characteristics make for a good alignment target? In their original report, Yu et al. [5] observed that ImageNet linear-probe accuracy correlates positively with generation performance, suggesting that global semantic representation is the driving factor. However, more recent work by Singh et al. [6] reports the opposite after profiling a wider range of encoders. Linear-probe accuracy does not in fact predict REPA gain, they argue, while a measure of the spatial structure of patch-token representations does. This suggests that generation performance for REPA may be driven by *local* feature geometry, rather than *global* semantic strength.

Which characteristics make for a good REPA encoder remains an open question, particularly outside the image setting. We pick up this thread in Chapter 4, where we analyse encoders for molecular generation and ask which of their properties predict REPA gain.

2.3 A primer on molecular generation

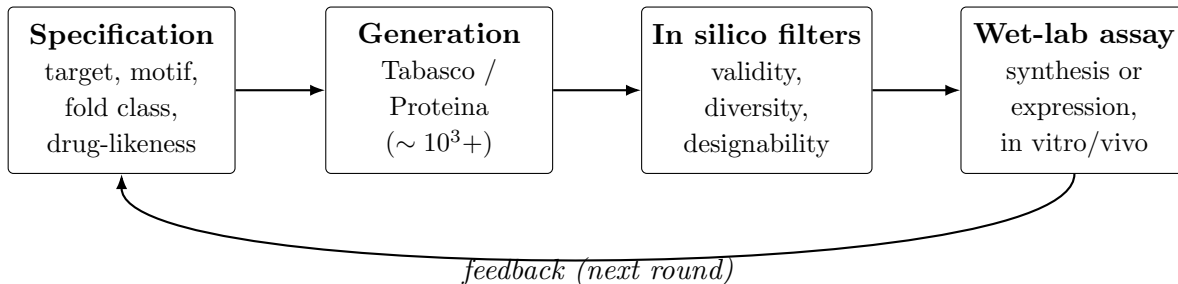


Figure 2.2: The de novo molecular design cycle. An upstream specification drives a generative model to produce a candidate pool, which is funnelled through in silico filters and then synthesised (or expressed) and assayed in the wet lab. Experimental results feed back into the next round of generation. Typical generation-step pool sizes are in the thousands to tens of thousands of candidates per round [17].

Having outlined the modelling paradigms at the heart of this thesis, we now turn to the molecular setting in which we apply them. To situate our work, we begin with a brief primer on the role generative models play within the broader de novo molecular design pipeline. Generative models are rarely intended to produce ready-to-use artefacts. Instead, they act as high-throughput in silico proposal engines in an iterative process of refinement and selection. This process, often called a *design campaign* [48], is summarised in Figure 2.2.

An upstream specification — such as a target binding site, a motif to scaffold, a fold class, or a drug-likeness profile — drives a generative model to produce a candidate pool: typically thousands to tens of thousands of candidates per round, in both small-molecule and protein-backbone domains [17]. The pool is funnelled through in silico filters: for small molecules these typically include structural validity and physical plausibility [49], drug-likeness [50], and target docking; for protein backbones, designability and structural diversity are the dominant considerations [13]. The select few survivors are synthesised in the laboratory or procured from vendor libraries (small molecules), or expressed in genetically engineered cell lines (proteins), and evaluated using in vitro or in vivo assays. Experimental results feed back into the next round of generation.

This pipeline framing is relevant to our discussion of evaluation metrics in Chapter 3. Several of the metrics we adopt — such as structural validity and designability — are not measures of intrinsic molecular quality, but proxies for specific in silico filter steps.

2.4 Tabasco and Proteina

The two models we study — Tabasco [14] for small molecules and Proteina [13] for protein backbones — are both non-equivariant transformer flow-matching generators. They are explicitly designed to scale with data and model size, rather than relying on structural priors imposed by model architecture. The rotational and translational symmetries expected of 3D molecules are intended to be learned from data and augmentation rather than imposed by construction.

Tabasco is motivated by the recent success of similar architectures in protein-structure generation like Proteina, setting it apart from equivariant message-passing small-molecule generators such as EQGAT-diff [43], MiDi [44], and SemlaFlow [45]. In turn, Proteina is motivated by the pair-bias attention mechanism introduced in AlphaFold 3 [39], which distinguishes it from SE(3)-equivariant backbone generators such as RFDiffusion [17], FrameFlow [41], and Genie [42]. Despite their architectural minimalism, both achieve performance comparable to equivariant baselines on standard structural-quality evaluations.

2.4.1 Small molecules and Tabasco

In medicinal chemistry, a *small molecule* is a drug-like compound at the sub-900-Dalton scale [46]. In silico, its structure is typically represented using two coupled features: a discrete *molecular graph* of typed atoms connected by typed bonds, and a continuous *conformer* placing those atoms in 3D Cartesian space. A flow-matching parameterisation must therefore handle both jointly.

Tabasco [14] represents each molecule as a sequence of per-atom tokens, each carrying an atom-type embedding and a 3D coordinate, but with no explicit bond decoder. This treats the molecule as a point cloud of typed atoms rather than a graph, sidestepping edge modelling at the cost of having to recover bond structure post hoc from interatomic distances. Tabasco is trained on GEOM-drugs [30], a public dataset of drug-like conformers.

This point-cloud-of-atoms parameterisation is one of several options in the literature, and each makes a different trade-off. *SMILES* and *SELFIES* are string encodings that work well with autoregressive models but discard the 3D conformer entirely; we encounter them later as the parameterisation employed by some candidate encoders we evaluate in Chapter 4. *3D voxel grids* preserve geometry, but lose the discrete graph and inflate the token count by an order of magnitude or more. *Equivariant point clouds*, used by generators such as EQGAT-diff [43], MiDi [44], and SemlaFlow [45], keep both graph and geometry, but pay the computational cost associated with SE(3)-equivariant layers. We refer the reader to Duval et al. [34] and Wang et al. [21] for broader surveys in this space.

2.4.2 Protein backbones and Proteina

A *protein* is a chain of amino-acid residues, ranging from around 50 to several thousands in length [47]; the largest known, titin, has approximately 34,000 residues. (Shorter chains are typically classified as *peptides* instead.) Each amino-acid residue contributes four *backbone* atoms (N, C $_{\alpha}$, C, O) whose interactions determine the chain’s *fold* geometry, and a *side chain* that determines its amino-acid identity.

For the structure-generation problem we study, it is customary to only model the backbone, with sequence identity recovered downstream using a separate *inverse-folding* model such as ProteinMPNN [23]. A further compression is the *CA-trace*, which uses only the set of C $_{\alpha}$ coordinates for every residue to represent the entire protein. This parameterisation retains enough information to recover fold topology and reduces computational cost relative to the more complete representation, which is typically referred to as *all-atom* modelling. Proteina [13] adopts this C $_{\alpha}$ parameterisation, representing each backbone as a sequence of per-residue tokens carrying continuous C $_{\alpha}$ Cartesian coordinates. The models reported by Geffner et al. are trained largely on high-confidence synthetic structures from the AlphaFold Database (AFDB) [29].

Once again, the CA-trace parameterisation is one of several techniques used in the literature to parameterise proteins in silico. *Sequence-only* models, such as the ESM lineage [22], operate on amino-acid identity strings. These models capture rich functional priors, but encode far less information on 3D structure. *Full-atom* representations close the gap to physical realism, but add substantial compute cost. *Equivariant* parameterisations, such as torsion angles and oriented residue frames, encode SE(3) symmetries by construction, and form the standard basis for the equivariant generators noted above. However, these architectures have historically bottlenecked scaling. We refer the reader to Notin et al. [51] for a broader survey of machine-learning approaches to protein design.

2.5 Open questions

This survey leaves two questions open that the rest of this report takes on:

- (i) *Does REPA work in the molecular domain?* Existing applications of REPA in molecular generation [7, 8] have not been ablated over encoder choice or evaluation-surface headroom, and protein backbone generation in particular remains unexplored.
- (ii) *When does REPA work?* The encoder-characterisation work surveyed in §2.2.1 has begun to ask which properties of an encoder predict REPA gain, but only within the image setting.

We take both questions up empirically in Chapters 4–6.

Chapter 3

Evaluating molecular generation

Chapter 4

Encoder targets and profiling

Chapter 5

REPA on small molecules

Chapter 6

REPA on protein backbones

Chapter 7

Conclusions

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Appendix A

Technical details

A.1 Placeholder section

TODO.